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Paleomagnetic study of Cambrian–Ordovician rocks in the Eastern Precordillera of Argentina: some constraints on the Andean uplift of this block

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Abstract

A paleomagnetic study on several carbonatic units (limestones to dolomites) of the Argentine Precordillera, spanning the Middle Cambrian to Arenig, is reported. Sampling was done at 23 sites (140 samples) in five localities of the Eastern Precordillera of San Juan province: Zonda (1), Juan Pobre (2) and La Flecha (3) creeks, Loma Redonda (4) and Cerro Pedernal de los Berros (5). Standard AF and thermal demagnetization indicated two different magnetic behaviors. Samples from localities 1 and 2 showed a single, post-tectonic magnetic component attributed to a Recent viscous or chemical remagnetization. Samples from localities 3 to 5 showed two magnetic components: a low unblocking temperature (350°C) magnetization coincident with that found at the previous localities and a high temperature one (up to 500°C). This second component was also found to be post-folding (post-Devonian?) and its in situ mean direction is Dec: 277.1°, Inc: 69.4, $\alpha 95$: 9.3°, $N=10$. This is not consistent with any expected direction for the study area between Carboniferous and Recent times. It is therefore inferred that the rocks carrying it underwent substantial tectonic rotation since its recording. Comparison with the magnetization carried by similar Early Ordovician limestones in the Central Precordillera and the San Rafael Block, and assigned to a Permian remagnetizing event, suggests that the same process affected the carbonatic rocks in the Eastern Precordillera and, therefore, the age of the component is likely Permian. Restoration of the in situ mean direction to the expected direction for a Permian remagnetization can be obtained by a 40° ccw rotation around an axis plunging 30° to N30°E. This is compatible with uplifting of the Eastern Precordillera as a series of quasi-rigid blocks along a major E dipping N to NE backthrust during the Andean orogeny. © 2000 Elsevier Science B.V. All rights reserved.

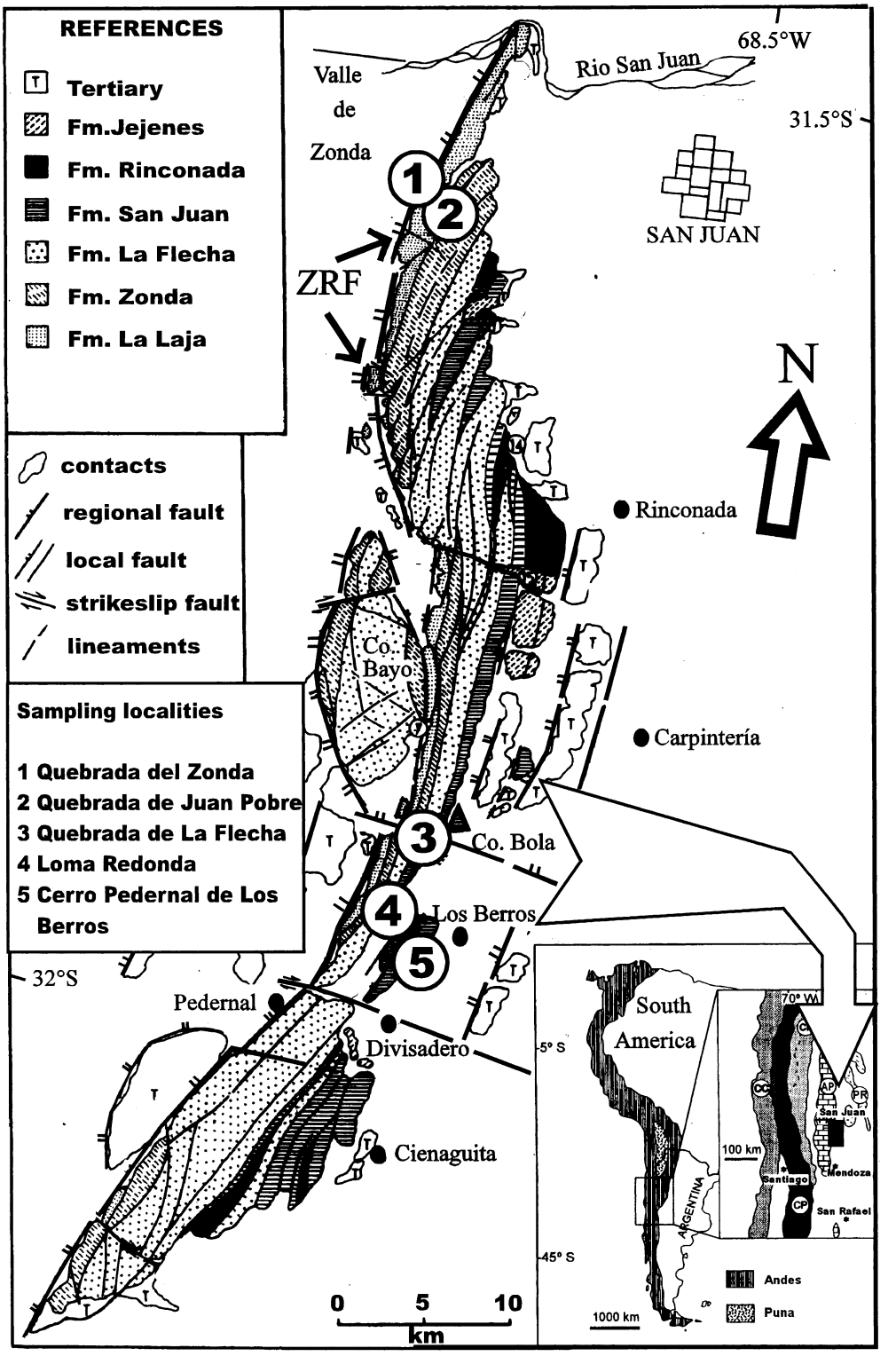
Keywords: paleomagnetism; Precordillera; Argentina; uplift; remagnetization; Andes

1. Introduction

The Argentine Precordillera, in the foothills of the Argentine Andes (Fig. 1), has recently been proposed to be an exotic fragment derived from Laurentia on the basis of biogeographic, stratigraphic and isotopic data (see Astini et al., 1995a; Benedetto et al., 1995; Mahlburg Kay et al., 1996). This model has been proved to be correct by paleomagnetism (Rapalini and Astini, 1998). These paleomagnetic results were obtained from

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REFERENCES
Tertiary
Fm.Jejenes
Fm. Rinconada
Fm. San Juan
Fm. La Flecha
Fm. Zonda
Fm. La Laja
contacts
regional fault
local fault
strikeslip fault
lineaments
Sampling localities
1 Quebrada del Zonda
2 Quebrada de Juan Pobre
3 Quebrada de La Flecha
4 Loma Redonda
5 Cerro Pedernal de Los Berros
68.5°W
31.5°S
32°S
Rio San Juan
Valle de Zonda
SAN JUAN
ZRF
N
Rinconada
Carpintería
Co. Bayo
Co. Bola
Los Berros
Pedernal
Divisadero
Cienaguita
0 5 10
km
South America
5° S
45° S
70° W
100 km
1000 km
Santiago
Mendoza
San Juan
San Rafael
ARGENTINA
Andes
Puna

Early Cambrian rocks exposed in the northern tip of the Argentine Precordillera. Previous paleomagnetic studies on samples from the very broad Middle Cambrian to Middle Ordovician carbonatic platform failed to obtain the primary remanence as most investigated rocks showed a pervasive Permian remagnetization related to a main orogenic event, the San Rafaelic phase (Rapalini and Tarling, 1993). This remagnetization apparently also affected other units in the Precordillera and Ordovician limestones in the San Rafael block, some 400 km south of the Argentine Precordillera (Truco and Rapalini, 1996). Several different tectonic models have been proposed for the mechanisms and kinematics involved in the departure of this terrane from Laurentia and its accretion to the southwestern margin of Gondwana (Dalla Salda et al., 1992; Thomas and Astini, 1996; Dalziel, 1997; Dickerson and Keller, 1998). The lack of paleomagnetic poles for the Argentine Precordillera in the interval Middle Cambrian–Late Ordovician precludes constraining its paleogeographic evolution and therefore testing the different tectonic models. In order to accomplish this, further paleomagnetic studies were carried out in Middle Cambrian to Early Ordovician carbonatic rocks exposed in the Eastern Precordillera of Argentina. Rapalini and Tarling (1993) had suggested that Permian remagnetization could be absent in the Eastern Precordillera. A paleomagnetic study on 23 sites is reported here. Although these rocks showed no sign of their primary remanence, a post-folding characteristic remanence, interpreted as the Permian remagnetization, was identified and determined at 10 sites. The displacement of this remanence direction by the uplift of the Eastern Precordillera block in Late Tertiary times permitted us to better define the kinematics of this uplift.

2. Geology and stratigraphy of the area

The Argentine Precordillera has been divided into Eastern, Central and Western sections. The Eastern Precordillera (Ortiz and Zambrano, 1981) is made up of three N–S trending ranges: from north to south, the Sierras de Villicum, Chica de Zonda and Pedernal , (the latter two are shown in Fig. 1). From a stratigraphic point of view it is characterized by the presence of a continuous succession of limestones and dolomites that span from the Early Cambrian (La Laja Fm) up to the Arenig (San Juan Fm). The stratigraphy and paleontologic content of these units have been studied widely (see for instance Baldis and Bordonaro, 1984; Baldis et al., 1982, 1984; Astini et al., 1995a,b; Peralta et al., 1995; Benedetto et al., 1995). The oldest unit exposed in the Eastern Precordillera is the La Laja Fm., the age of which spans from the upper Lower Cambrian (*Olenellus* zone) to the upper Middle Cambrian (*Bollaspidella* zone). It reaches an approximate maximum thickness of 1000 m, in a composite section (Baldis and Bordonaro, 1984). It shows a broad range of lithologic types, although its dominant sediments are subtidal mudstones and wackestones. Bordonaro (1980) subdivided this formation into four members. Only the upper two members: Rivadavia and Juan Pobre, of Middle Cambrian age, were sampled for this study. On top of these is the Zonda Fm, composed of some 300 m of dolomites and calcareous dolomites of early diagenetic origin. It has been assigned to a peritidal environment (Astini et al., 1995b). Its age is constrained by the fossil assemblages found in its bounding units to the lower Upper Cambrian. On top of the Zonda Fm lies the La Flecha Fm with around 500 m of limestones and dolomites with very abundant stromatolites and trombolites, and a fossil fauna indicative of an Upper Cambrian age. On top of it is the La Silla Fm, which encompasses the Cambrian–Ordovician boundary. It is composed mainly of limestones with few dolomitic levels intercalated. Its maximum thickness reaches 350 m. The youngest and broadest distributed calcareous unit is the San Juan Fm. It consists of 300 m of massive grey limestones and thin-laminated light grey limestones and marls. Its

Fig. 1. Geologic map of the Sierra Chica de Zonda and Cerro Pedernal, Eastern Precordillera. The paleomagnetic sampling localities are indicated by numbers. ZRF: Zonda regional fault. Modified from Baldis and Bordonaro (1984).

paleontologic content indicates an age ranging from Late Tremadoc to Early Llanvirn, its top being apparently diachronous.

From a structural point of view, the Argentine Precordillera is a Late Tertiary peri-Andean thin-skinned fold-and-thrust belt that extends over 400 km in length between the provinces of Mendoza and La Rioja. Although its main lithologic constituents are of Paleozoic age, the present-day features are strongly controlled by the Andean tectonic regime (Allmendinger et al., 1990; Jordan et al., 1983a,b). The Argentine Precordillera corresponds to a latitudinal stripe where the Nazca plate is characterized by a subhorizontal subduction regime, which has produced the migration of the orogenic front toward the east up to the Sierras Pampeanas in the provinces of Córdoba and San Luis. The most important structural features of the Argentine Precordillera are submeridional thrust faults, that are mainly responsible for the tectonic shortening of 50% or more according to different models (von Gosen, 1992; Introcaso et al., 1992; Ramos, 1995). The main decollement zones occur in the Early Paleozoic limestones (Baldis and Chebli, 1969), which accounts for the lack of exposure of the Precambrian basement. Most of these thrusts are east vergent, except for those at the Eastern Precordillera which are backthrusts vergent to the west. According to Zapata and Allmendinger (1996), the structures of the Eastern Precordillera are governed by the pre-Andean structural fabric of the basement. The tectonic activity in the Eastern Precordillera apparently also involves the basement (Smalley et al., 1993), which suggests a thick-skinned model for this block. Uplift of the Eastern Precordillera apparently started only about 3 Ma ago and continues today (Smalley et al., 1993; Zapata and Allmendinger, 1996).

3. Sampling and experimental results

140 samples were collected with a gasoline-powered portable drilling machine at 23 sites. These sites were distributed in five localities in the Eastern Precordillera (Figs. 1 and 2) as follows: sites JS1 to JS4 (La Laja Fm) and JS7 to JS10

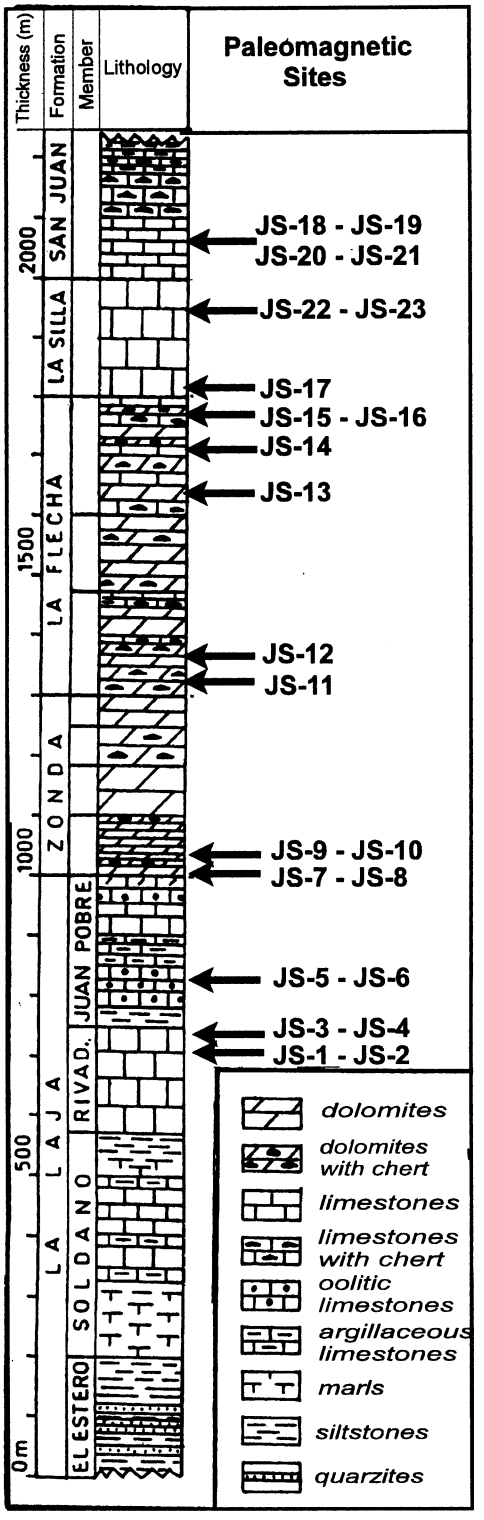


Fig. 2. Stratigraphic distribution of paleomagnetic sites.

(Zonda Fm) at locality 1 (Quebrada del Zonda); sites JS5 and JS6 (La Laja Fm) at locality 2 (Quebrada de Juan Pobre); sites JS11 to JS16 (La Flecha Fm) and JS 17 (La Silla Fm) at locality 3 (Quebrada de La Flecha); sites JS18 to JS 21 (San Juan Fm) at locality 4 (Loma Redonda); and sites JS22 and JS23 (La Silla Fm) at locality 5 (Cerro Pedernal de los Berros). Whenever possible samples were oriented by sun and magnetic compass.

Two pilot specimens per site were submitted to detailed AF and thermal demagnetization. Measurements of intensity and direction of remanence were done with a 2G DC cryogenic magnetometer, AF demagnetization was performed with a three-axis static degausser attached to the magnetometer. Thermal demagnetization was done with a standard Schonstedt TSD-1 oven. Stages for stepwise demagnetization were as follows: 2, 4, 6, 8, 10, 15, 20, 25, 30, 40, 50, 60, 80, 100 and 120 mT and 100, 150, 200, 250, 300, 350, 400, 450 and 500°C. No significant remanence remained over 500°C. After this pilot study thermal demagnetization was selected as the most resolute treatment. Therefore all remaining samples were thermally demagnetized in steps of 150, 300, 350, 400, 440, 470 and 500°C.

The paleomagnetic behavior of the samples could be divided into two groups. Samples from sites 1 to 10 (localities 1 and 2) only showed the presence of a single magnetic component (Fig. 3) which could be well determined by principal component analysis (Kirshvink, 1980) and trended toward the origin of coordinates, with both AF and thermal demagnetization. The direction of this component (A) was coincident with the expected geocentric axial dipolar field direction at these localities (Fig. 4). This and the single normal polarity found in all processed samples of these sites strongly suggest that these rocks carry a secondary Recent overprint of probably viscous or chemical origin. Due to these results only two or three samples were fully processed from sites 1 to 10. Samples from sites 11 to 23 showed two magnetic components (Fig. 5). These were much better identified by thermal demagnetization. A low temperature component was found in almost all samples. This component was completely erased at 300–350°C. Its direction is consistent with component A found at localities 1 and 2 (Fig. 4). Therefore, it is also interpreted as a Recent secondary overprint. A higher temperature component (B) was unblocked between 350 and 500°C (Fig. 5). This component was also defined in most sites by principal component analysis (Kirshvink, 1980); MAD values lower than 15° were considered acceptable due to the low intensity of the remanence, although over 50% showed values $<10^{\circ}$. At sites 11 and 12, a joint analysis of remanence directions and remagnetization circles (McFadden and McElhinny, 1988) was performed to define the mean site directions. Despite the fact that component B was apparently present in all sites, sites 12, 18 and 19 were excluded from the analysis due to the poor statistical parameters of their mean directions. The remaining sites showed moderate to good directional consistency (Table 1). All sites except JS-20 (Fig. 5J) presented a direction of positive inclination (reverse polarity). The unblocking temperature spectra of samples from this site are almost identical to those of the remaining ones, suggesting that the same characteristic magnetization but with opposite polarity was recorded. The in situ site mean directions are very well grouped but become inconsistent after bedding correction is applied (Fig. 6, Table 1). This clearly indicates that component B is also a post-folding magnetization. Later considerations will show, however, that B is not entirely post-tectonic.

The paleomagnetic behavior of the samples from localities 3 to 5 resembles those from the San Juan limestones studied by Rapalini and Tarling (1993). These authors determined the presence of a secondary magnetization in several outcrops of the San Juan limestones in Central Precordillera. This magnetization was assigned to a Permian remagnetizing event associated with the San Rafaelic tectonic phase. The carrier of the remanence was found to be (Ti poor?) magnetite of an apparent authigenic origin, in a similar fashion to the well-defined remagnetized limestones of Eastern North America (McCabe and Elmore, 1989). Detailed rock-magnetic studies of the rocks studied here are still under way (Berquo et al., in preparation), however, some information already

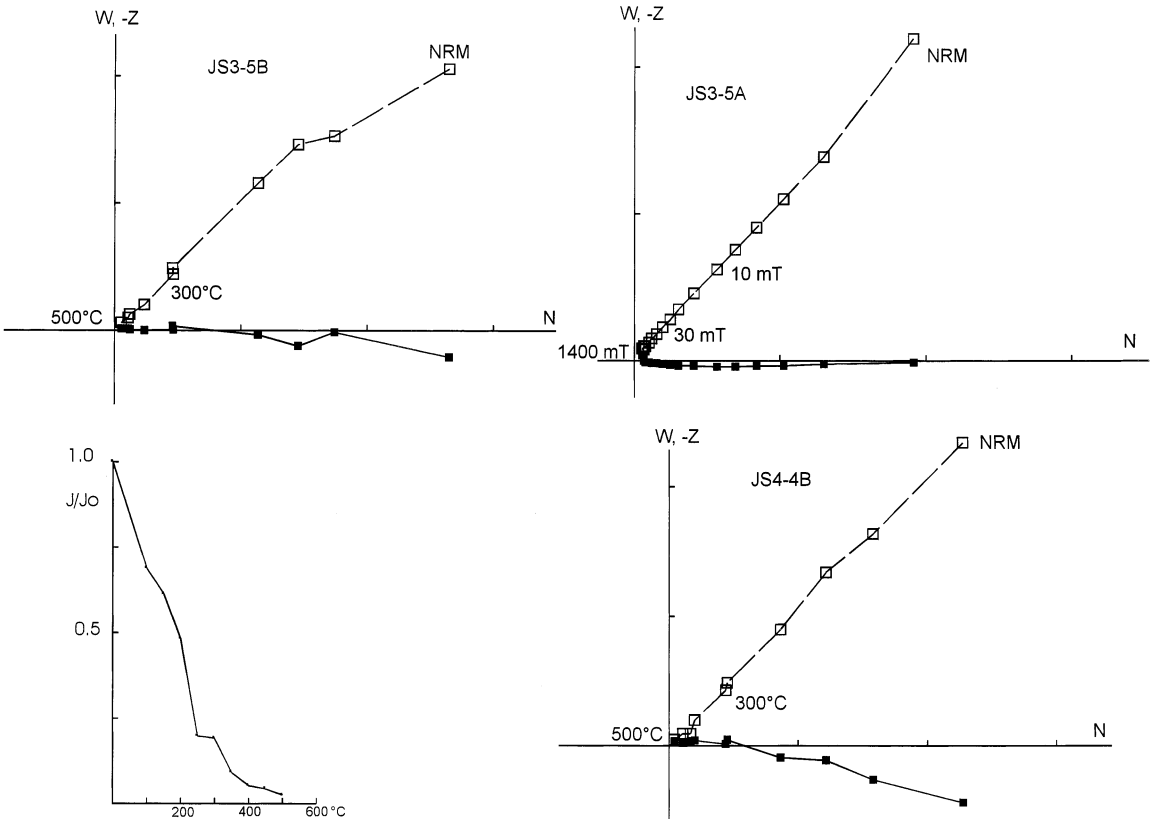


Fig. 3. Characteristic magnetic behavior of samples from localities 1 and 2 submitted to AF or thermal demagnetization. Note the presence of a single magnetic component. The demagnetization curve corresponds to sample JS3-5B. Solid symbols: projection on horizontal plane.

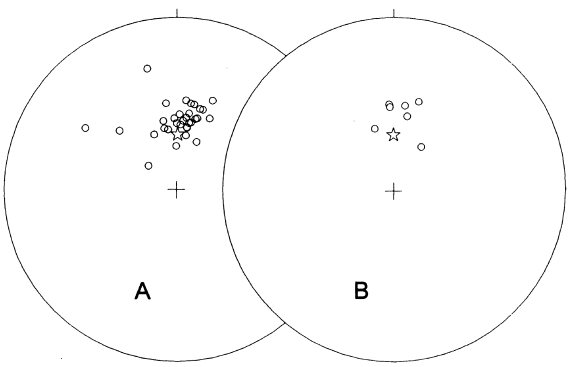


Fig. 4. (A) Characteristic remanence directions per sample from localities 1 and 2 (sites JS1 to JS10). (B) Mean site directions of low temperature (300–350°C) component from localities 3, 4 and 5. Star indicates the direction of the axial geocentric dipole magnetic field at the sampling area.

obtained also suggests that the carrier of component B is most likely titano-magnetite. This is supported by the unblocking temperatures around 500°C and hysteresis parameters compatible with this mineralogy. Fig. 7 shows the typical Hcr/Hc vs. Jrs/Js diagram (Day et al., 1977) for a sample per site of this study. While all samples except one from localities 1 and 2 fall in the MD sector of the diagram, supporting a viscous origin for their remanence, samples from the other localities (in which component B was found) have a wider distribution with some in the MD field, while others fall in or close to the sector of the remagnetized limestones from Eastern North America. The latter mainly correspond to the Early Ordovician San Juan limestone (sites JS18 to JS22), which is the same formation in which the Permian remag-

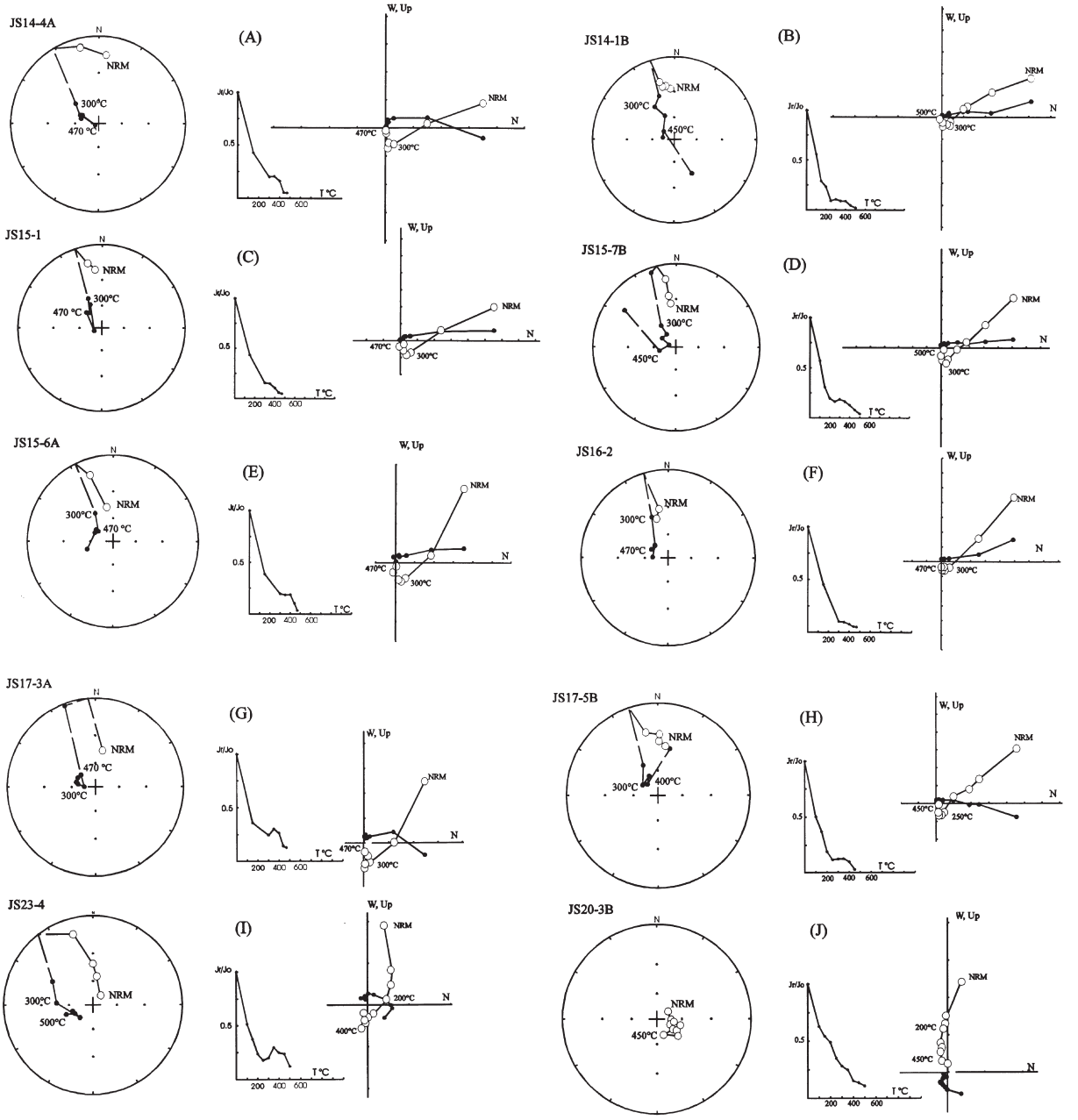


Fig. 5. Demagnetization behavior of representative samples from localities 3, 4 and 5 submitted to thermal cleaning. Note the presence of two magnetic components, one (component A) with unblocking temperatures around 300°C and a second one (component B) with unblocking temperatures about 450–500°C. See discussion in the text.

netizing event was found in Central Precordillera (Rapalini and Tarling, 1993). This suggests some differences in magnetic mineralogical content between this formation and the remaining units. However, component B was observed and defined not only in this formation but in the La Flecha

Table 1
Site mean remanence directions of component B. 'Strike' and 'Dip' apply for bedding. 'Dec*' and 'Inc*' denote declination and inclination after bedding correction

Site	N	Dec	Inc	α95	k	Strike	Dip	Dec*	Inc*
JS-11	5	237.6°	64.9°	18.2°	23.7	4°	88°	109.7°	21.9°
JS-13	4	199.9°	60.8°	11.6°	63.6	18°	83°	137.4°	7.0°
JS-14	5	294.4°	69.7°	10.3°	55.7	4°	89°	86.6°	20.0°
JS-15	6	310.1°	70.4°	6.2°	117.5	4°	89°	82.1°	16.7°
JS-16	5	301.5°	70.1°	12.0°	41.4	4°	89°	84.4°	18.6°
JS-17	4	290.8°	78.3°	14.4°	41.8	4°	89°	90.6°	12.2°
JS-20	5	126.8°	−62.7°	20.2°	15.2	44°	77°	318.3°	−40.0°
JS-21	4	276.6°	56.5°	22.5°	17.6	44°	77°	159.2°	38.0°
JS-22	5	290.0°	55.2°	15.0°	27.0	230°	65°	303.4°	−5.8°
JS-23	6	264.5°	69.4°	7.6°	78.8	220°	75°	295.4°	0.2°
Mean sites in situ	10	277.1°	69.4°	9.3°	27.8				
Mean sites after correction	10	111.2°	19.6°	17.3°	8.7				

and La Silla Formations too. Hysteresis loops showed a dominance of the paramagnetic or diamagnetic signal, with a very subordinate ferromagnetic contribution. This is due to the very small amount of ferromagnetic grains, which is also evidenced in the very low NRM intensities (10^{-2} mA/m). The fact that nine out of 10 site directions correspond to a reverse field is also consistent with the exclusive reverse polarity found in the San Juan limestones in Central Precordillera. Economy of hypothesis suggests that it is very likely that component B found at localities 3 to 5 in the La Flecha, La Silla and San Juan Formations is the same Permian remagnetization that was found in the Central Precordillera by Rapalini and Tarling (1993). The normal polarity remanence isolated in a single site may suggest that in this location remagnetization processes were still active in the Late Permian. The interpretation of a Permian remagnetization is also consistent with structural data, as shown below. If true, it proves that this event affected different carbonatic units and extended into the Eastern Precordillera.

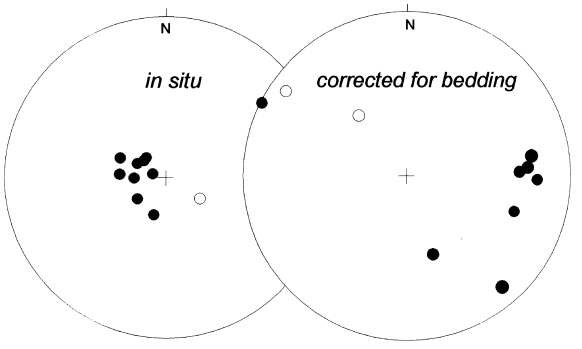


Fig. 6. Mean site directions of component B from localities 3, 4 and 5, in situ and after bedding correction. Note the post-folding nature of this component.

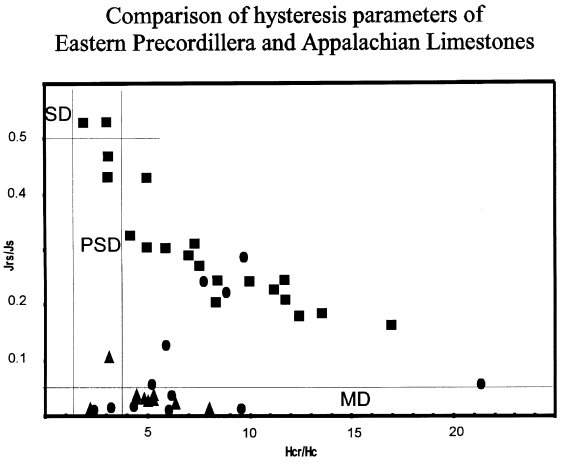


Fig. 7. Hcr/Hc versus Jrs/Js for a representative sample per site of the studied rocks. Triangles: samples from localities 1 and 2, ovals: samples from localities 3 to 5, squares: samples from remagnetized limestones from Eastern North America (Jackson, 1990). SD, single domain; PSD, pseudo-single domain; MD, multidomain fields (Day et al., 1977).

4. Structural implications

Component B is clearly a post-folding magnetization. However, its mean direction, Dec: 277.1°, Inc: 69.4°, α95: 9.3°, $N=10$, is far removed from any expected post-Devonian direction for the sampling area. This indicates that the remanence directions were affected by post-magnetization rotations. The rocks under study underwent at least two major tectonic phases (Baraldo et al., 1996). One occurred in pre-Carboniferous times (Late Devonian?) and produced folding of the carbonatic succession at a mesoscopic scale. This phase was responsible for folding, for example, of the outcrops sampled at locality 4 (Loma Redonda), in which sites 18–19 and 20–21 belong to opposite limbs of an anticline. A later tectonic phase (Andean) was initiated in the Late Tertiary and apparently continues in present times. This compressive Andean activity produced mainly NNE backthrust with associated folding. This is represented by the Zonda regional fault: a main backthrust that marks the western boundary of the Villicum, Zonda and Pedernal ranges (Fig. 1). This faulting affected the Late Tertiary and even Quaternary deposits that were tilted toward the east. According to the previous discussion, the removal of B direction from its expected position must be due to the Andean uplift of this sector of the Eastern Precordillera by backthrust faulting. Rotation of B back to its expected position can therefore be used to constrain the uplift kinematics of this block. The directional consistency of sites from different localities suggests that a large area of the Southeastern Precordillera, i.e. that located between the La Flecha and Rio del Agua creeks, behaved as a single rigid block during its uplift, without important internal deformation. The Zonda regional fault has been interpreted as a high angle thrust fault that involves the basement of the Eastern Precordillera, suggesting that this is a thick-skinned range (Ramos, 1995; Gardini, 1993). According to Gardini (1993) and Ramos (personal communication), the depth to the top of the basement should be on the order of 12 km at the Zonda range, although von Gosen and Buggisch (1992) have proposed a much shallower (around 5 km) basement. This does not agree with evidence from drills at the Matagusanos valley, where over 6000 m of post-Ordovician sediments were drilled (Gardini, 1993). The uplift of the Eastern Precordillera would mainly consist of a passive rotation of the block along the Zonda regional fault. The sedimentary covers were also affected by several shallower second-order thrust faults that did not involve the basement. With such simplified geometric constraints, uplift of the Eastern Precordillera along the main Zonda fault would imply a clockwise (looking north) rotation of the block around a NNE axis, i.e. the rotation pole to restore B to the expected position should lie on the backthrust plane. Fig. 8 shows the in situ position of the mean site direction for component B together with the expected direction for the Permian remagnetization (SJ, Rapalini and Tarling, 1993) and for the geocentric axial dipole (T), which is the approximate Late Tertiary direction. B is rotated to coincide with SJ by 40° ccw rotation around an axis trending N30°E and plunging 30° toward the NNE. The azimuth of the Zonda regional fault south of Quebrada de La Flecha is about N30°E. If, instead, B is to be

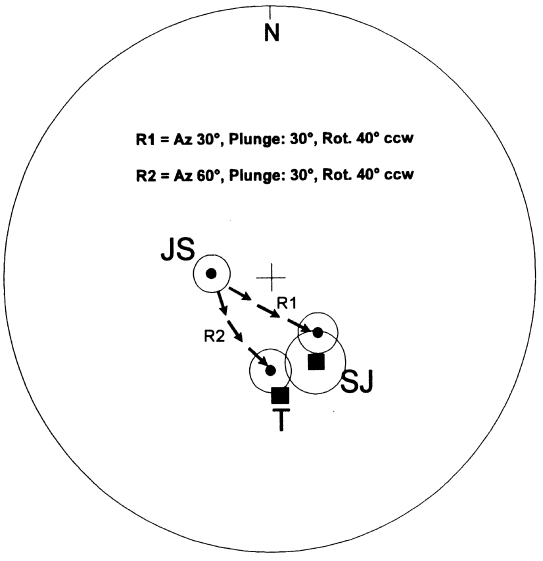


Fig. 8. Site mean direction of component B (JS) in situ and after being rotated according to the rotation axes and values indicated to make it consistent with the expected directions for the San Rafaelic remagnetization (SJ) and a Late Tertiary overprint (T). See details in the text.

rotated to coincide with the Late Tertiary expected position (T in Fig. 8), rotation must take place around an axis trending N60°E. There is no obvious correlation of this direction with any fault plane in the region. This is supportive of the previous interpretation of component B as a Permian remagnetization. The Zonda and Pedernal ranges are flanked by Late Tertiary deposits on the east. These deposits are systematically tilted between 25 and 40° ESE. This is consistent with the 40° value of rotation inferred from the paleomagnetic data. This implies that the fold sampled at locality 4 was originally overturned. This is not surprising since the Late Devonian (?) Chañic orogenic phase was very intense (Ramos et al., 1986; Baraldo et al., 1996).

Different throw values have been calculated for the Zonda fault. Cardinali (1997) computed a throw of around 2 km in the Sierra Chica de Zonda, north from the study area. On the other hand, balanced cross-sections presented by Ramos (1995) and Gardini (1993) portray a vertical displacement of around 6 and 10 km, respectively, for the basement at approximately the same location. Cardinali (1999) also agrees with these values. These higher values are more in accordance with a 40° rotation of the upper block along the Zonda fault. A rough calculation suggests that around 7 to 8 km of vertical displacement would result from the paleomagnetically determined rotation along a subcircular thrust fault with a decollement horizon located 14 km below the surface. In particular, the balanced cross-section of Gardini (1993) requires a 40° eastward tilt of the Early Paleozoic deposits, in perfect accordance with the paleomagnetic data. Movements on the Zonda fault must also have occurred prior to the Late Tertiary (Gardini, 1993; Cardinali, 1999; Ramos, personal communication), since this last uplift could only account for around 5 km of displacement (Cardinali, 1999). Although quasi-rigid rotation on the main backthrust apparently governed the uplift kinematics of the Eastern Precordillera, the paleomagnetically determined values are only applicable to the block limited by the La Flecha and Rio del Agua creeks. Several E–W normal and/or strike–slip faults have been determined in the Eastern Precordillera (Fig. 1). It is likely that these faults separate different crustal blocks with probably slightly to moderately different values of rotation. Paleomagnetic determinations on some of these other blocks could better constrain the kinematics of the uplift of the Eastern Precordillera.

5. Conclusions

The paleomagnetic study of Middle Cambrian to Early Ordovician carbonates exposed in the Eastern Precordillera of Argentina could not find evidence of the primary remanence in any of the 23 studied sites. While samples from sites near the Quebrada del Zonda only showed the presence of a magnetic component coincident with the geomagnetic field direction, samples from the remaining sites at Quebrada de La Flecha and nearby localities showed the presence of a higher unblocking temperature component (B) apparently carried by (Ti poor?) magnetite. Similar demagnetization behavior as well as structural considerations suggest that component B corresponds to the Permian remagnetizing event associated with the San Rafaelic tectonic phase and detected in similar limestones in Central Precordillera. Component B could be determined at 10 sites, and its in situ mean direction is Dec: 277.1°, Inc: 69.4°, $\alpha 95$: 9.3°, $N=10$. Application of bedding correction indicates a post-folding (post-Devonian?) character for this component. This overall mean direction is far removed from any expected post-Devonian direction. It is concluded that the remanence underwent a significant rotation due to the uplift of the Eastern Precordillera in Late Tertiary to Quaternary times, although part of the rotation could have happened earlier. Restoration of the B mean direction to the expected direction for the San Rafaelic remagnetization indicates that this rotation, of 40° magnitude, took place around an axis trending N30°E and plunging 30° north. This axis is parallel to the strike of the Zonda regional fault, the principal backthrust that marks the western boundary of the Eastern Precordillera. It is finally inferred that the uplift of this sector of the Eastern Precordillera took place as a quasi-rigid body, since dispersion of remanence directions among sites is small.

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